

Zhaodong Liu

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EDUCATION

New York University Shanghai

Bachelor of Science; Double Major: Computer Science, Mathematics, Cum Laude

Shanghai, China

Sep 2022-May 2026

Cumulative GPA: **3.803/4.0** | CS Major GPA: **3.946/4.0** | Math Major GPA: **3.818/4.0**

- **Study-away Experience:** NYU New York, Fall 2024 and Spring 2025
- **Relevant Coursework:** Multivariable Calculus, Honors Linear Algebra, Probability & Statistics, Honors Numerical Analysis, Discrete Mathematics, Analysis, Ordinary Differential Equations, Data Structures, Basic Algorithms, Operating Systems, Introduction to Databases, Machine Learning, Natural Language Processing, Stochastic Processes
- **Honors:** Dean's List (Academic Years: 2022-2025), Founders' Day Award (Annual Distinguished Student Award)

Carnegie Mellon University

Master of Science; Computational Data Science

Pittsburgh, PA

Aug 2026-Dec 2027 (expected)

RESEARCH EXPERIENCE

Research Assistant, New York University Shanghai

May 2025-May 2026

Generative Retrieval for Multimodal Recommendation System

Advisor: Prof. Hongyi Wen, Department of Computer Science, Data Science and Engineering

- Developed a multimodal framework to predict user behavioral patterns on Amazon datasets.
- Designed and evaluated model architectures, implemented quantization methods of VQ-VAE, RQ-VAE, OPQ, to capture discrete behavior patterns, achieving 47.2% performance improvement (from 0.0339 to 0.0499).
- Implemented BPE-inspired partitioning tokenization, improving sequence coherence and representation quality.

Research Assistant, New York University Shanghai

May 2024-Aug 2024

Efficient Driving Route Optimization Considering Traffic Light Timings

Advisor: Prof. Zhibin Chen, Department of Computer Science, Data Science and Engineering

- Built a traffic-aware driving route optimization model under real-world map and constraints on SUMO platform.
- Simulated model on real traffic signal datasets, reducing travel time compared to baseline routing algorithms.
- Applied energy consumption models for EV and gasoline vehicles, balancing energy efficiency and travel time.

SELECTED PROJECTS

Finetuning on Pretrained LM for Efficient Retrosynthesis of Chemical Reactions Sep 2024-Dec 2024

Center for Data Science, New York University

- Developed predictive models for chemical reaction retrosynthesis using SMILES notation.
- Applied an LLM (LlaSMol-Llama2) for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (from 0.588 to 0.742) while reducing reliance on expensive laboratory data.
- Employed fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to optimize training efficiency.

Fantasy Sports League Database

Oct 2024-Dec 2024

Tandon School of Engineering, New York University

- Designed a 13-table relational database in MySQL supporting multi-league sports (Football, Basketball, Soccer) with user authentication, draft scheduling, roster management, trading systems, and 390+ sample records.
- Implemented ACID-compliant transaction management, automated triggers, and stored procedures to enforce referential integrity and achieve efficient query performance through proper indexing.
- Built a Python backend for user authentication, data querying, and business logic; developed an HTML/CSS frontend for visual operations, integrating the full stack with the MySQL database.

Collaborative Piano

Mar 2025-May 2025

Tandon School of Engineering, New York University

- Built a networked real-time piano application in Java enabling two users to perform together, featuring synchronized audio playback, integrated chat, and session recording with precise timestamps.
- Implemented TCP socket networking with thread concurrency and Java Sound API supporting multiple instrument timbres including synthesized waveforms and sampled piano audio.

- Applied Graphics2D and AffineTransform for an animated metronome, integrating GUI graphics, file I/O, and socket networking into a cohesive real-time multimedia system.

WORK EXPERIENCES

Data Engineer Intern, Data Strategy Department, McDonald's China Dec 2025-Apr 2026

- Developed and validated scalable database schemas and enterprise-level data maps.
- Designed a similarity recommendation algorithm based on menu composition and user historical interactions.
- Implemented an AI agent that significantly enhanced data panel classification and visualization.

LEADERSHIP & ACTIVITIES

Treasurer, PEER-Pal Club, New York University Shanghai May 2023-Aug 2024

- Managed financial operations and annual budgeting for student-led educational initiatives.
- Facilitated letter exchanges and Q&A sessions with high school students from Hunan Province.

Trombone Section Leader, Chamber Orchestra, New York University Shanghai Sep 2022-May 2026

- Directed the trombone section, coordinated rehearsals, and oversaw instrument storage logistics.

DISTINCTIONS

Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention May 2023

- Preprocessed large datasets to remove anomalies and ensure data quality.
- Built predictive models using methods including Bayesian regression for regional market data analysis.

SKILLS AND OTHERS

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- **Programming:** Python, Java, C, SQL, Stata, R
 - **Frameworks & Tools:** Git, PyTorch, Pandas, NumPy, LaTeX, Markdown
 - **Languages:** Chinese (Native), English (Fluent)